

L4A: Introduction to Data Abstraction

CS1101S: Programming Methodology

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Outline

- ▶ Data abstraction (2.1)

<https://sourceacademy.nus.edu.sg/sicpy/2.1>

- ▶ Case study: rational numbers (2.1.1)

<https://sourceacademy.nus.edu.sg/sicpy/2.1.1>

- ▶ Representing sequences with pairs (2.2.1)

<https://sourceacademy.nus.edu.sg/sicpy/2.2.1>

Announcements

- ▶ RA1 this Friday!
- ▶ Contest "Beautifully Built" closes soon!
- ▶ Unit 1 Survey open, closes Friday 23:59
 - In Google Form
 - Get 200 XP

Where Are We Now?

► Module overview

- Unit 1 — Functional abstraction: SICP Chapter 1
- Unit 2 — Data abstraction: SICP Chapter 2
- Unit 3 — State: SICP Chapter 3
- Unit 4 — Beyond: SICP Chapters 4 & 5

► New language: Python §2

- See language documentation at https://docs.sourceacademy.org/python/python_2/
- Source Academy Playground is using Python §2 from now on <https://sourceacademy.nus.edu.sg/playground>

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Types of Values in Python

- ▶ Numbers: 1, -5.6, 0.5e-157
- ▶ Boolean values: True, False
- ▶ Strings: `"this is a string"`
- ▶ Functions: `lambda x: x + 1`
- ▶ Some others (imported from modules):
`heart, make_point(0.5, 0.25)`
- ▶ Today: **pairs** and **None**

Recall first_denomination function in cc (L3)

```
def first_denomination(kinds_of_coins):  
    return (5 if kinds_of_coins == 1  
            else 10 if kinds_of_coins == 2  
            else 20 if kinds_of_coins == 3  
            else 50 if kinds_of_coins == 4  
            else 100 if kinds_of_coins == 5  
            else 0)
```



- This first_denomination function “stores” five values and “retrieves” them when given the proper index

Data Structure Example

- ▶ `first_denomination` is a data structure
- ▶ Construct the data structure
 - using function definition

```
def first_denomination(kinds): ...
```

- ▶ Select components of the data structure
 - using function application

```
... first_denomination(3) ...
```


Data Structures in Mathematics

- ▶ They are **everywhere**: tuples, sets, matrices, etc.
- ▶ What is the simplest data structure possible?
- ▶ A **pair**
 - **Constructed** in math using tuple notation, e.g. $(0.5, 0.25)$
 - **Selected** in math using a pattern:
 - Let p be (x, y) , for some x and y ... (and now use x and y)

Pairs so far — Points in Curve Missions

```
# Construct:  
p = make_point(0.5, 0.25)  
  
# Select:  
x = x_of(p)  
y = y_of(p)
```

We can Define the Point Abstraction (2.1.3)

```
def make_point(x, y):  
    return lambda component:  
        x if component == 0 else y  
  
def x_of(p):  
    return p(0)  
  
def y_of(p):  
    return p(1)
```



Using More Generic Names

```
def pair(x, y):  
    return lambda component:  
        x if component == 0 else y  
  
def head(p):  
    return p(0)  
  
def tail(p):  
    return p(1)
```



Another way to define pair, head, tail

```
pair = lambda x, y: lambda f: f(x, y)

head = lambda p: p(lambda x, y: x)

tail = lambda p: p(lambda x, y: y)
```

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Case Study: Rational Numbers (2.1.1)

- ▶ What is a rational number?
 - A pair, consisting of a denominator and a numerator
- ▶ In Python:

```
def make_rat(n, d):  
    return pair(n, d)  
  
def numer(x):  
    return head(x)  
  
def denom(x):  
    return tail(x)
```

Addition and Subtraction of Rational Numbers

```
def add_rat(x, y):  
    return make_rat( numer(x) * denom(y) +  
                     numer(y) * denom(x),  
                     denom(x) * denom(y))
```

```
def sub_rat(x, y):  
    return make_rat( numer(x) * denom(y) -  
                     numer(y) * denom(x),  
                     denom(x) * denom(y))
```


Multiplication and Division of Rational Numbers

```
def mul_rat(x, y):  
    return make_rat( numer(x) * numer(y),  
                     denom(x) * denom(y))
```

```
def div_rat(x, y):  
    return make_rat( numer(x) * denom(y),  
                     denom(x) * numer(y))
```

Equality of Rational Numbers

► First attempt:

```
def equal_rat(x, y):  
    return (numer(x) == numer(y) and  
            denom(x) == denom(y))
```

► Second attempt:

```
def equal_rat(x, y):  
    return (numer(x) * denom(y) ==  
            numer(y) * denom(x))
```

Printing Rational Numbers

```
def rat_to_string(x):  
    return str( numer(x) ) +  
           " / " +  
           str( denom(x) )
```

Playing with Rational Numbers

```
print(rat_to_string(add_rat(one_half, one_third)))
```



► 5 / 6

```
print(rat_to_string(mul_rat(one_half, one_third)))
```



► 1 / 6

```
print(rat_to_string(add_rat(one_third, one_third)))
```



► 6 / 9

Making Reduced Rational Numbers

- Compute the greatest common divisor (GCD) of two numbers using Euclid's algorithm

```
def gcd(a, b):  
    return a if b == 0 else gcd(b, a % b)  
  
def make_rat(n, d):  
    g = gcd(n, d)  
    return pair(n // g, d // g)
```

Playing with Rational Numbers Again

```
print(rat_to_string(add_rat(one_half, one_third)))
```



► 5 / 6

```
print(rat_to_string(mul_rat(one_half, one_third)))
```



► 1 / 6

```
print(rat_to_string(add_rat(one_third, one_third)))
```



► 2 / 3

Data Abstraction Barriers

Programs that use rational numbers

Rational numbers in problem domain

`add_rat` `sub_rat` ...

Rational numbers as numerators and denominators

`make_rat` `numer` `denom`

Rational numbers as pairs

`pair` `head` `tail`

However pairs are implemented

Summary of Case Study on Rational Numbers

- ▶ Pairs can be used to represent rational numbers
- ▶ Operations are implemented using constructor and selector functions
- ▶ `...rat...` functions hides the internal representation of the data
 - Implementation details remain invisible to the user of the library
 - Provides a higher-level abstraction

Outline

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- ▶ Representing sequences with pairs (2.2.1)
<https://sourceacademy.nus.edu.sg/sicpy/2.2.1>

Representing Sequences with Pairs: Motivation

- ▶ Want to put the coin denominations 100, 50, 20, 10, 5 in a “sequence” constructed from pairs

```
first_denomination = pair(...100...50...20...10...5...)
```

- ▶ Many possible ways, examples:
 - `pair(pair(100, 50), pair(20, pair(10, 5)))`
 - `pair(100, pair(pair(50, 20), pair(10, 5)))`
 - `pair(pair(pair(100, 50), pair(20, 10)), 5)`
 - `pair(100, pair(50, pair(20, pair(10, 5))))`
- ▶ Different ways of representations require different ways of retrieval

Idea: Introduce Some Discipline

► Principle

- Make sure that `head(p)` always has the **data**, and `tail(p)` always has the **remaining elements**

► Example:

```
denoms = pair(100, pair(50, pair(20, pair(10, 5))))
```

► `head(denoms) ⇒ 100`

► `tail(denoms) ⇒ pair(50, pair(20, pair(10, 5)))`

Special Case

► What if

```
denoms = pair(10, 5)
```

► Then the program

```
rest = tail(denoms)
```

gives us a value 5, not the remaining **sequence of elements**

Idea: Introduce a Base Case

- ▶ How to represent the empty sequence?
 - It doesn't really matter!
- ▶ In Python, we use the special value `None` to represent a sequence of no elements

The first_denomination! using None

```
first_denomination = pair(100,  
                           pair(50,  
                                pair(20,  
                                     pair(10,  
                                          pair(5,  
                                               None))))))
```

Linked List Discipline in Python

- ▶ Definition: A **linked list** is either None or a pair whose tail is a linked list
- ▶ We will call it a `l1ist` (**li-list**) for short

```
# Examples:
my_l1ist = None

your_l1ist = pair(8, None)

first_denomination =
    pair(100,
        pair(50,
            pair(20, pair(10, pair(5, None))))))
```

Retrieving Data from a Linked List

► Example:

```
denoms =  
    pair(100, pair(50, pair(20, pair(10, pair(5, None)))))
```

`head(denoms)` $\Rightarrow$ 100

`head(tail(denoms))` $\Rightarrow$ 50

`head(tail(tail(denoms)))` $\Rightarrow$ 20

`head(tail(tail(tail(denoms))))` $\Rightarrow$ 10

`head(tail(tail(tail(tail(denoms)))))` $\Rightarrow$ 5

`tail(tail(tail(tail(tail(denoms)))))` $\Rightarrow$ None

LINKED LISTS Functions in (Python §2)

- ▶ `pair(x, y)` — returns pair made of `x` and `y`
- ▶ `is_pair(p)` — returns `True` iff `p` is a pair
- ▶ `None` — represents an empty linked list
- ▶ `is_none(xs)` — returns `True` iff `xs` is the empty linked list `None`
- ▶ `head(p)` — returns the head (first component) of the pair `p`
- ▶ `tail(p)` — returns the tail (second component) of the pair `p`
- ▶ `llist(x1, ..., xn)` — returns a linked list whose first element is `x1`, second element is `x2`, etc. and last element is `xn`
- ▶ ...

Box Notation

- In box notation

`print(pair(x, y))` is printed as `[x, y]`

Empty linked lists are printed as `None`

- Example:

```
print(pair(1, pair(2, pair(3, None))))
```



is printed as

`[1, [2, [3, None]]]`

Linked List Notation

- ▶ Same as box notation, but any sub-structure that is a llist is nicely formatted and printed as `llist(...)`
- ▶ Use predeclared function `display_list(x)` to show `x` in llist notation
- ▶ Example:

```
print_llist(  
    pair(pair(pair(7, 8), pair(1, pair(2, None))), 6))
```



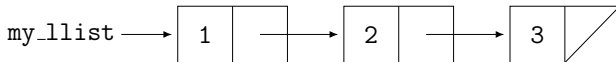
prints

```
[llist([7, 8], 1, 2), 6]
```

Box-and-Pointer Diagrams

- Box-and-pointer diagrams are graphical representations of data structures made of pairs

Example: `my_llist = pair(1, pair(2, pair(3, None)))`



- Data Visualizer tool generates such diagrams in Playground
- Use the `draw_data` primitive function

```
draw_data(pair(1, pair(2, pair(3, None))))
```



Check-In I4A

Time for a Sound Check(-In)!

Look for Check-In I4A in the Source Academy!

Error Reporting

- ▶ The functions that query the structure of linked lists have expectations for their arguments:
 - `head(xs)`: expects `xs` is a pair
 - `tail(xs)`: expects `xs` is a pair
- ▶ Otherwise, a nice error message gets printed

Length of a Linked List

► Definition:

The length of the empty linked list is 0, and the length of a non-empty linked list is one more than the length of its tail

► Examples:

- Length of None is 0
- Length of `pair(10, None)` is 1
- Length of `pair(10, pair(20, pair(30, None)))` is 3

Computing the Length of a Linked List

```
def length(xs):  
    return (0  
            if is_none(xs)  
            else 1 + length(tail(xs)))
```



- Does it give rise to a recursive or iterative process?

Computing the Length of a Linked List

► Iterative version:

```
def length_iter(xs):  
    def len(ys, counted_so_far):  
        return (counted_so_far  
                if is_none(ys)  
                else len(tail(ys), counted_so_far+1))  
    return len(xs, 0)
```



Summary

- ▶ Data structures are everywhere
- ▶ Pairs
- ▶ Case study: rational numbers
- ▶ Linked List discipline
- ▶ LINKED LISTS functions in Python §2
- ▶ Box notation, List notation, and Box-and-pointer diagrams